

ADEBANJI OLUWATIMILEYIN ADELOWO

Mathematical Engineer | Scientific Machine Learning Engineer

☎ +39 388 751 5527 ✉ adelowooluwatimileyin@gmail.com 🌐 adebanjioluwatimileyin.github.io

🌐 linkedin.com/in/adebanjioluwatimileyin 🐙 github.com/AdebanjiAdelowo

EDUCATION

- M.Sc. Mathematical Engineering | University of L'Aquila, Italy** 2018 – 2021
- Thesis: *Lower Bounds on the Mix Norm of Passive Scalars Advection by Incompressible Enstrophy-Constrained Flows*
 - Coursework: PDEs, Functional Analysis, Numerical Analysis, Finite Element Methods, Scientific Computing, Optimization, Machine Learning, Fluid Dynamics
- B.Sc. Mathematics | Obafemi Awolowo University, Nigeria** 2014 – 2018
- Thesis: *Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem*
 - Coursework: Real Analysis, Complex Analysis, Topology, Numerical Analysis, PDEs, Probability, Mathematical Modeling

RESEARCH EXPERIENCE

- Master's Thesis Research | University of L'Aquila, Italy** 2020 – 2021
- Derived analytical lower bounds on mix norms for passive scalars under energy and enstrophy constraints using variational methods and functional inequalities
 - Identified optimal velocity fields accelerating mixing; implemented pseudo-spectral simulations in Python porting Gautam Iyer's MATLAB code
- Undergraduate Thesis Research | Obafemi Awolowo University, Nigeria** 2017 – 2018
- Applied Galerkin FEM to the one-dimensional Stefan (free-boundary) problem; developed algorithms for phase boundary tracking in Python

PROFESSIONAL EXPERIENCE

- Data Scientist | Intuos Srl, L'Aquila, Italy** Dec 2024 – Sep 2025
- Developed **ML Classifiers** (Random Forest, LGBMClassifier) for real-time flight phase identification and **FFT-based signal processing pipelines** for engine RPM estimation and anomaly detection
 - Built REST APIs, automated telemetry pipelines, and **operational dashboards** for flight performance analysis
- Machine Learning Engineer | N-and Italia Srl, Pognano, Italy** Apr 2023 – Nov 2024
- Developed predictive ML models for vending machine delivery failure forecasting from large-scale transaction and telemetry logs
 - Built KPI dashboards and performed statistical analysis of system performance and operational anomalies
- Operational Data Analyst | eResult Srl, Cesena, Italy** Jan – Mar 2023
- Optimized SQL queries and automated reporting pipelines for business intelligence systems
- Digital Engineering Intern | Bridgestone EMIA, Rome, Italy** Apr – Oct 2022
- Improved deepcopy efficiency on GeometryMap half-edge data structures in tire cross-section FEA meshing pipelines
 - Developed geometric algorithms for cubic Bézier curve closest-point computation using scalar minimization
- Machine Learning Intern | Fellowship.AI, San Francisco (Remote)** May – Aug 2022
- Implemented **InsetGAN architecture** for whole-body human image generation; deployed inference service to a Swift iOS app via FastAPI

HONORS AND AWARDS

- University of L'Aquila International Scholarship | University of L'Aquila** 2018 – 2021
- Professor S.S. Okoya Prize | Excellence in Applied Mathematics, Obafemi Awolowo University** 2018
- MTN Foundation Scholarship | Merit-based undergraduate scholarship** 2014 – 2018
- PTDF Scholarship | Petroleum Technology Development Fund, Nigeria** 2014 – 2018
- Murli T. Chellaram Foundation Scholarship | Academic excellence award** 2014 – 2018

TEACHING AND MENTORSHIP

Founder & President, Inter-University Mathematics Club | Nigeria

2016 – 2018

- Established cross-university mentorship network; organized seminars on mathematics, data science, AI, and research careers

Academic Director, NAMSS | Peer Tutor, Obafemi Awolowo University, Nigeria

2015 – 2018

- Coordinated workshops and invited industry and academic speakers; tutored undergraduates in analysis and mathematical reasoning

TECHNICAL SKILLS

Programming: Python, SQL, MATLAB, C++, R, $\text{\LaTeX}$

ML & AI: PyTorch, TensorFlow, scikit-learn, LangChain, OpenAI API

Engineering & Data: FastAPI, Docker, PostgreSQL, IBM DB2, Git, FFT, finite elements, pseudo-spectral methods

Visualization & Cloud: Matplotlib, Plotly, Power BI, React/Vite, AWS, Google Cloud

SELECTED PROJECTS

INTUOS Flight Data Monitoring Dashboard | Full-stack aviation analytics platform; FastAPI + IBM DB2 backend, React/Vite frontend. Real-time telemetry, flight phase classification, engine metrics.

Aircraft Engine RPM Estimation via Audio DSP | Raspberry Pi system using FFT on microphone audio for RPM extraction; validated on Tecnam P92 / Rotax 912.

LLM Engineering Portfolio | Medical RAG Assistant, Enterprise NL-to-SQL, Financial Sentiment Distillation. Stack: LangChain, OpenAI, Pinecone.

Image Inpainting | TV-regularized Douglas–Rachford optimization for variational image restoration and inverse imaging.

Roof Segmentation | U-Net for aerial image segmentation with convolutional feature extraction and data augmentation.

Vending Machine Failure Prediction | Predictive ML for delivery failure forecasting from telemetry logs across 150+ machines.

Subsurface Skin Scattering | Dipole BSSRDF renderer for all 6 Fitzpatrick skin types in pure PyTorch; analytically derived and measured R_{hemi} across skin tones. Stack: PyTorch, NumPy

NeRF vs TensorRF | Pure-PyTorch NeRF and TensorRF trained on a synthetic scene; TensorRF achieves 17.2 dB PSNR / 0.834 SSIM vs NeRF's 14.2 dB / 0.770. Stack: PyTorch, MPS

Differentiable PBR Rendering | Cook–Torrance differentiable renderer optimising albedo, surface normals, roughness, and metallic maps from multi-view observations via photometric loss. Stack: PyTorch, NumPy

3D Liver Segmentation — Abdominal CT | 3D U-Net trained end-to-end on MSD Task03 liver CT volumes (131 cases). Foreground-biased patch sampling, combined soft-Dice + BCE loss, cosine LR annealing, Gaussian sliding-window inference. Best val Dice **0.9886**, HD95 ≤ 1 mm after 200 epochs on Kaggle T4. Stack: PyTorch, MONAI, NumPy